

Vehicle-to-Grid(V2G) in Thailand

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TGC-EMC

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AGENDA

- What is V2G?
- V2G IN Thailand: TGC-EMC
- International best Practices(V2X-Business Cases)
- Stakeholders' perspectives on V2G
- V2G assessment model
- Simulation results from selected use cases

WHAT IS V2G?

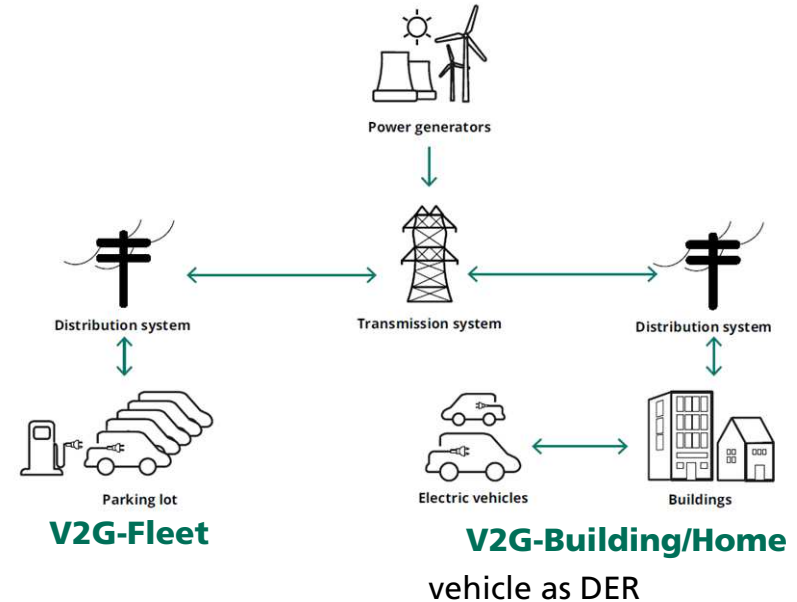
Definition:

The term vehicle-grid integration (VGI) covers the systems in which electric vehicles (EVs) can communicate with the power grid and, in this way, render unidirectional or bidirectional services to it. The technology is also often called vehicle-to-grid or V2G (Elia 2022).

V1G: 'managed (or smart) charging', the charging of EVs is controlled (e.g., delaying EV charging to avoid times of high-power demand).

V2G: EVs can return electricity to the grid in a controlled manner.

Additional services: Frequency regulation, Demand Response, Energy Storage and Backup etc.

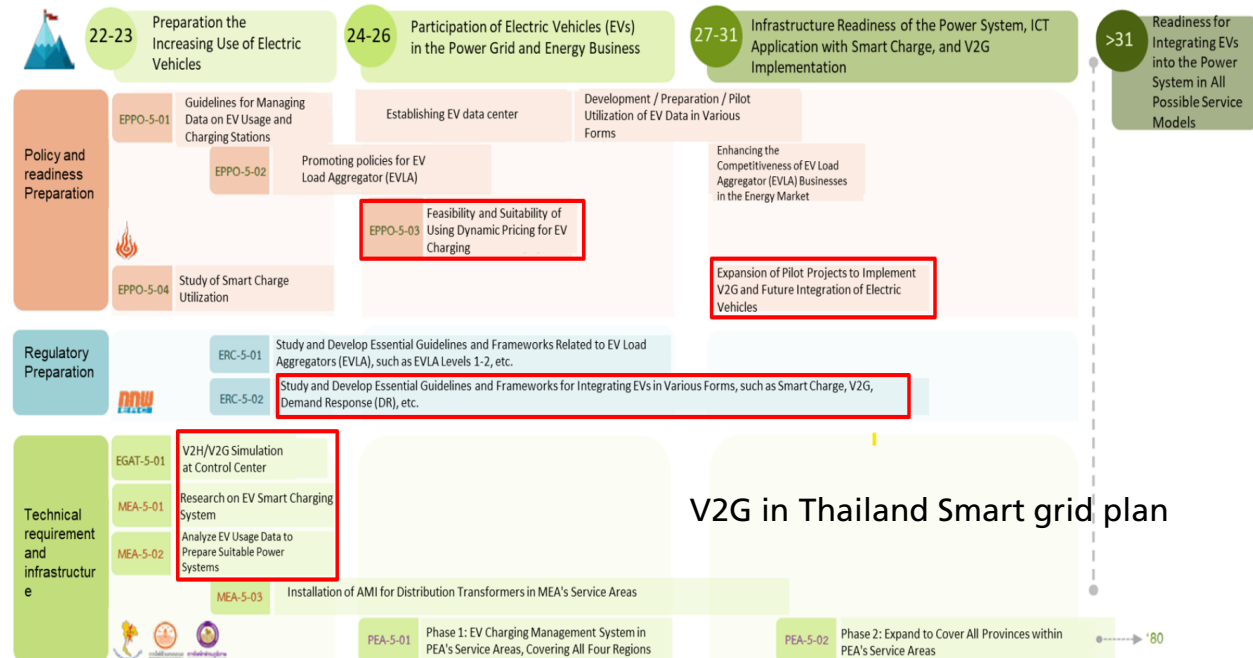


Ala et al. (2020).

V2G IN THAILAND: TGC-EMC

What are we doing in TGC-EMC?

- Supporting V2G implementation in Thailand.
- Knowledge sharing on Global V2G best practices.
- Facilitating Stakeholder exchange.
- Assessing and quantifying the potential benefits and evaluating different use cases.



INTERNATIONAL BEST PRACTICES(V2X-BUSINESS CASES)

Building level (Local level)	• Self consumption • Emergency Backup power supply • Local microgrid • Supply to home/building appliance
Market service	• Market connected dynamic tariff • Trading/Charging-as-a-service (aggregation) • CO2 optimized tariff • Arbitrage
Grid-side service	• Demand Response Services • Frequency regulation • Load Balancing (Peak Shaving) • Redispatch (TSO/DSO) • Local grid services (DSO) • Reactive power provision (DSO/TSO)



<https://www.thai-german-cooperation.info/resource/vehicle-to-grid-for-thailand-a-global-v2g-best-practices/>

STAKEHOLDERS' PERSPECTIVES ON V2G



“V2G implementation in Thailand requires collaborative actions, consumer awareness, market and regulatory reforms”

Issues and Gap	Regulatory	Technical	Market	Consumer
Missing collaboration among stakeholders	✖		✖	
Lack of charging Infrastructure	✖			✖
No national grid code for V2G	✖	✖		
Data sharing among stakeholder and with service providers	✖			✖
Standardization (technical and safety)	✖	✖		
Lack of Incentive and Subsidies	✖		✖	
Lack of awareness about V2G potential			✖	✖
Lack of suitable business model			✖	✖
Concern about battery life, safety and range anxiety		✖		✖
EV Conversion standardization	✖	✖		
Smart-metering and smart-grid	✖	✖		
Competitive energy price for V2G	✖		✖	✖

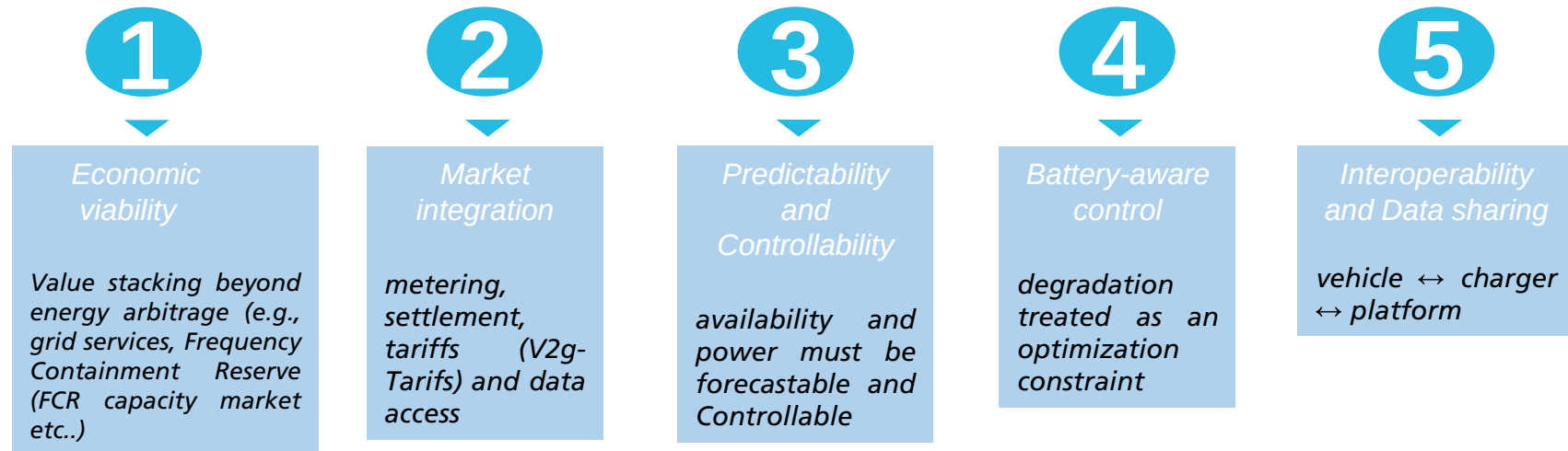
STAKEHOLDERS' PERSPECTIVES ON V2G

- **Smart charging and demand response** are important for managing EV integration. However, regulation, technical standards, and investment conditions need to improve.
- EV charging should be practical, user-friendly, and well integrated with mobility needs and renewable electricity.
- **Vehicle-to-grid (V2G) deployment requires clear regulations, interoperable standards, consumer acceptance, and attractive incentives.**
- Growing EV and battery markets create opportunities, but sufficient charging infrastructure, **viable business models, and policy support are needed.**
- EV charging will increasingly affect local distribution grids. **Grid planning, flexibility mechanisms,** and coordinated charging are essential to prevent congestion

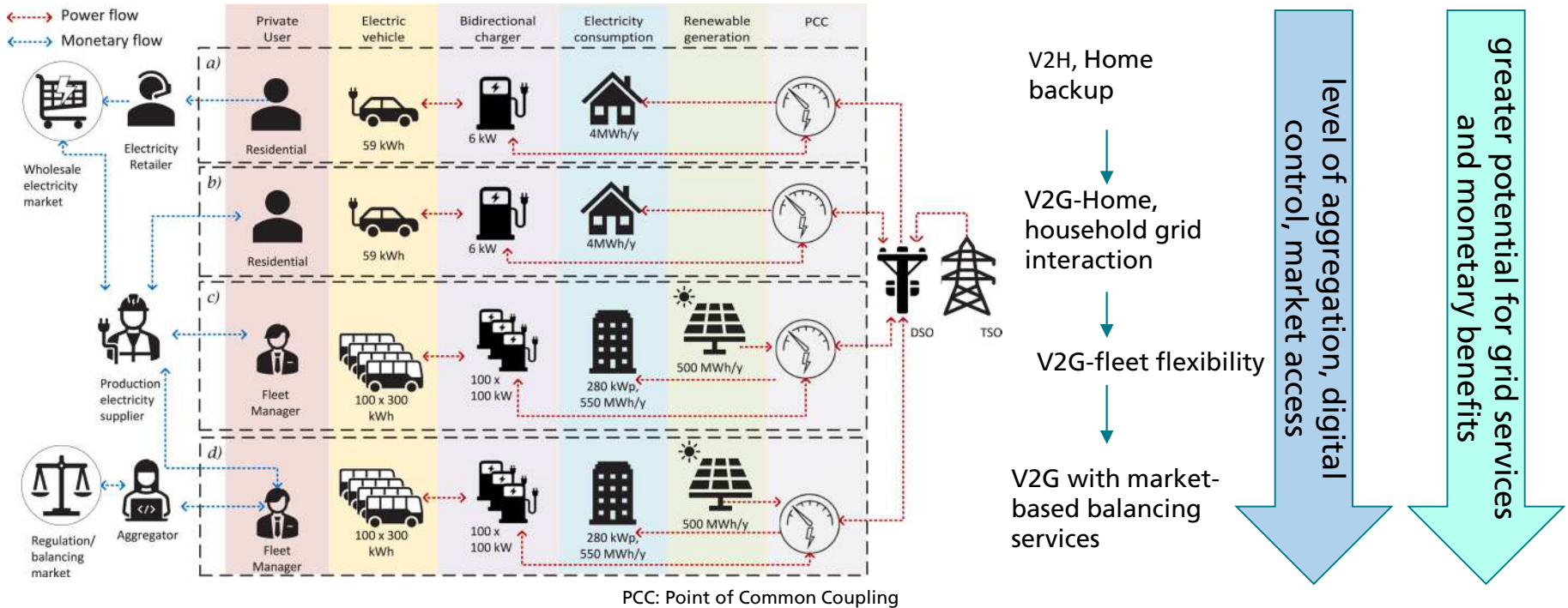


KEY FINDINGS ON V2G IMPLEMENTATION IN THAILAND

In Thailand, V2G is not primarily a technology, hardware, or bidirectional energy-flow challenge. The key challenge is to establish a **viable business case** that brings together stakeholders and end-users into a **functioning V2G ecosystem**.



V2G BUSINESS MODELS



V2G BUSINESS MODELS

PCC: Point of Common Coupling

Thanks !!!

Q&A

Contact